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SET-B #15

# JOSÉ RIZAL UNIVERSITY

80 SHAW BOULEVARD  
MANDALUYONG CITY

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## EXAMINATION BOOK

NAME

ANTHONY S. MINARES

DATE

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INSTRUCTOR

DR. RODOLFO RAGA

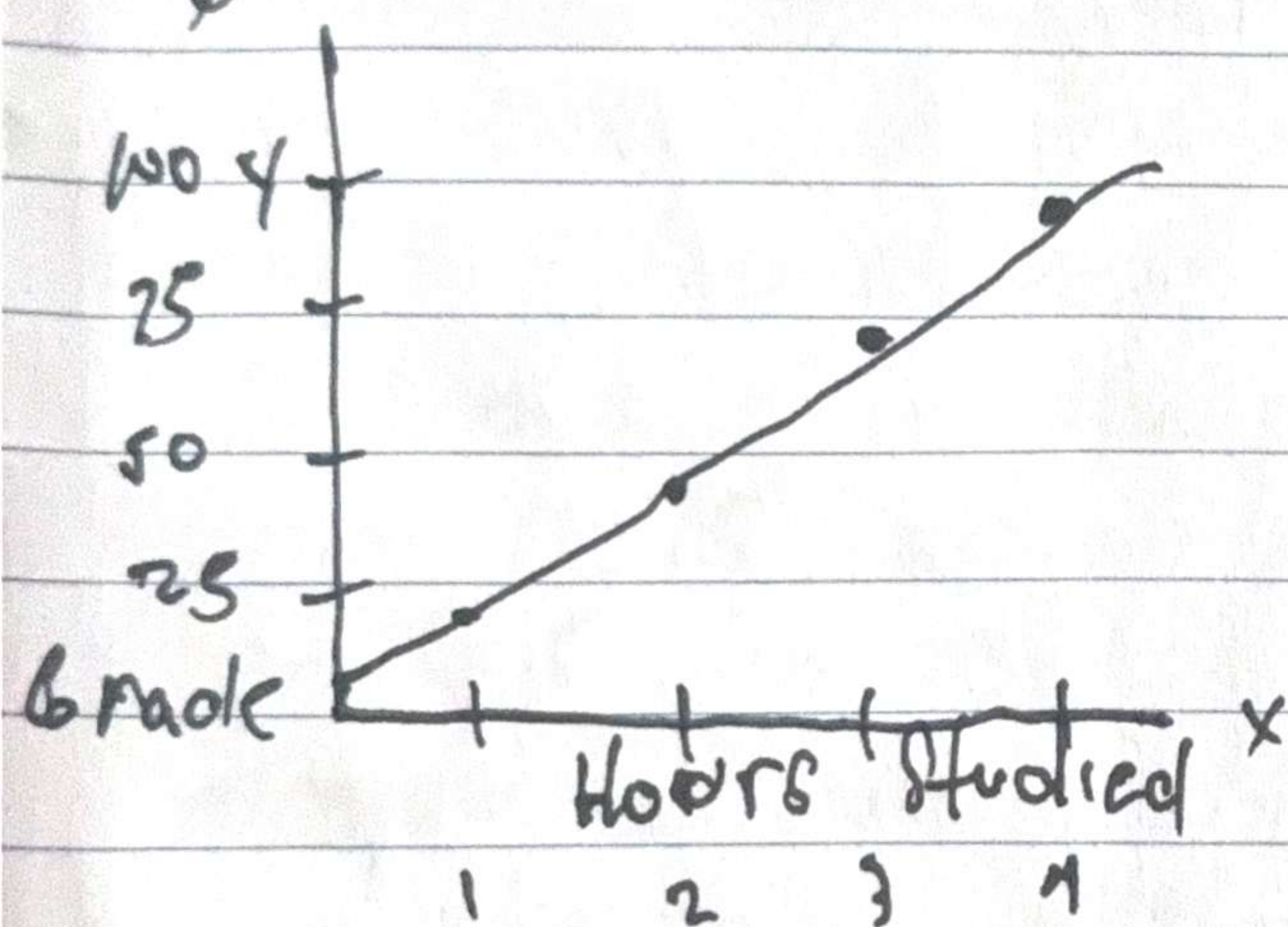
SUBJECT

ITCSOL - BOIi

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EXAMINATION BOOK

## TEST 1:

1. Linear regression is really not a viable option in predicting a "Yes/No" outcomes as it follows a straight line in a sense that the example "Hours Studied" with a categorical Yes or No that is represented by 1 or 0, hence if you utilize Linear Regression, the predicted value may exceed 1 or perhaps lower than 0.



In hours studied as shown on the graph that the more hours you spent studying the higher the chance you will pass. Although, since it follows a straight line it will come to a point (~~that the grade~~) that the model won't be able to determine if the value is a passing mark, which is represented by 1 or a failing mark, which is 0 without utilizing sigmoid curve model.

opposite

3.) It follows the ~~same~~ logic about the example given regarding the prediction if a patient is diabetic, where it will be detrimental to the patient if the prediction is yes, the patient is diabetic but actually don't. In contrast to that, the bank utilizes Recall by computing the number of people predicted correctly that they will pay over the same number of people predicted correctly that they will pay and those people predicted who will pay but actually won't to ensure if the person applying for a loan meet the requirement in terms

~~(Recall =  $\frac{TP}{TP + FN}$ )~~

of credit score and the formula being followed is written below:

$$\text{Recall} = \frac{TP}{TP + FN}$$

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## TEST II

| stocks | Type        | Reviews | Returns | Popular |
|--------|-------------|---------|---------|---------|
| 1      | FURNITURE   | 3.7     | 359     | No      |
| 38     | KITCHENWARE | 4.8     | 24      | yes     |
| 721    | ELECTRONICS | 2.4     | 3       | yes     |
| 46     | CAR MEKS    | 1.3     | 21      | No      |
| 77     | ELECTRONICS | 2.6     | 513     | No      |

| Attribute | Data Type   |             |
|-----------|-------------|-------------|
| Stocks    | Numerical   | Independent |
| Type      | Categorical | Independent |
| Reviews   | Numerical   | Independent |
| Returns   | Numerical   | Independent |
| Popular   | Categorical | target      |

The main benefit that an e-commerce company (~~that they~~) that they can get out of in predicting the target attribute, is they can anticipate whether there is a demand on a specific product and if ~~that~~ it is they can get enough supply. On the other hand if it is not popular enough they can reduce the stocks in order to allocate funds on popular products.

I believe that the four predictors are enough and do have enough weighted value in making an accurate and real-world prediction as the stocks might have a low value but may get a mediocre review but has a lot

or returns, and the target variable is still not popular, hence getting the model to ~~explore~~ <sup>utilize</sup> a real-world dataset is the best way to start in making or creating a reliable model.

The main benefit that an entrepreneur  
 can get from the market is that they can  
 get out of an organization the best return  
 they can and stop it when they like it  
 because of a specific product and  
 it is then that they can get out of it  
 in the other way it is not possible  
 when you have the stock in order to  
 allocate funds in different products.

I believe that the far greater one  
is the one that is not written down  
but is in the mind of the person  
who is writing it. I think that is the  
one that is the most important one.